

# Bangladesh University of Business and Technology



## LAB REPORT

**Experiment No** : 03  
**Course Title** : Algorithms Lab  
**Course Code** : CSE 232  
**Experiment Name:** Lab Assignment 03

### Submitted By:

**Name** : Kawsar Hoque  
**ID No** : 20255103270  
**Intake** : 55  
**Section** : 7  
**Program** : B.Sc. Engg. in CSE

### Submitted To:

**Name** : Sumona Yeasmin  
*(Lecturer)*  
**Dept. of** CSE  
**Bangladesh University of Business  
& Technology**

**Date of Submission** : 17/06/2026

## 1) Merge Sort

```
Start here x mergesort270.cpp x
1  #include <iostream>
2  using namespace std;
3
4  void merge(int A[], int l, int m, int h)
5  {
6      int n1 = m - l + 1;
7      int n2 = h - m;
8
9      int L[n1], R[n2];
10
11     for (int i = 0; i < n1; i++)
12         L[i] = A[l + i];
13     for (int j = 0; j < n2; j++)
14         R[j] = A[m + 1 + j];
15
16     int i = 0, j = 0, k = l;
17
18     while (i < n1 && j < n2)
19     {
20         if (L[i] <= R[j])
21         {
22             A[k] = L[i];
23             i++;
24         }
25         else
26         {
27             A[k] = R[j];
28             j++;
29         }
30         k++;
31     }
32     while (i < n1)
33     {
34         A[k] = L[i];
35         i++;
36         k++;
37     }
38     while (j < n2)
39     {
40         A[k] = R[j];
41         j++;
42         k++;
43     }
44 }
45
46 void mergesort(int A[], int l, int h)
47 {
48     if (l < h)
49     {
50         int m = (l + h) / 2;
51         mergesort(A, l, m);
52         mergesort(A, m + 1, h);
53         merge(A, l, m, h);
54     }
55 }
56
57 int main()
58 {
59     int n;
60     cout << "Size of array: ";
61     cin >> n;
62
63     int arr[n];
64
65     cout << "Enter your array: ";
66     for (int i = 0; i < n; i++)
67     {
68         cin >> arr[i];
69     }
70
71     if (n > 0)
72     {
73         mergesort(arr, 0, n - 1);
74     }
75
76     cout << "Sorted array: ";
77     for (int i = 0; i < n; i++)
78     {
79         cout << arr[i] << " ";
80     }
81     cout << endl;
82
83     return 0;
84 }
```

Output:

```
*C:\Users\Kawsar Hoque\Desl x + v
Size of array: 5
Enter your array: 8 3 9 5 4
Sorted array: 3 4 5 8 9

Process returned 0 (0x0)   execution time : 19.484 s
Press any key to continue.
```

## 2) Quick Sort - Pivot as First Element (Using For Loop)

```
1  #include <iostream>
2  using namespace std;
3
4  int partition(int A[], int l, int h)
5  {
6      int pivot = A[l];
7      int j = l;
8
9      for (int i = l + 1; i <= h; i++)
10     {
11         if (A[i] < pivot)
12         {
13             j++;
14             swap(A[i], A[j]);
15         }
16     }
17     swap(A[l], A[j]);
18     return j;
19 }
20
21 void quickSort(int A[], int l, int h)
22 {
23     if (l < h)
24     {
25         int pivot = partition(A, l, h);
26         quickSort(A, l, pivot - 1);
27         quickSort(A, pivot + 1, h);
28     }
29 }
30
31 int main()
32 {
33     int n;
34     cin >> n;
35     int arr[n];
36     for (int i = 0; i < n; i++)
37         cin >> arr[i];
38     if (n > 0)
39         quickSort(arr, 0, n - 1);
40     for (int i = 0; i < n; i++)
41         cout << arr[i] << " ";
42     return 0;
43 }
```

```
"C:\Users\Kawsar Hoque\Desl  X  +  v  -  □  X
5
9 8 7 6 5
5 6 7 8 9
Process returned 0 (0x0)   execution time : 21.705 s
Press any key to continue.
```

### 3) Quick Sort - Pivot as First Element (Using While Loop)

```
1  #include <iostream>
2  using namespace std;
3
4  int partition(int A[], int l, int h)
5  {
6      int i = l;
7      int j = h;
8      int pivot = A[l];
9
10     while (i < j)
11     {
12         while (i <= h && A[i] <= pivot)
13             i++;
14         while (A[j] > pivot)
15             j--;
16         if (i < j)
17             swap(A[i], A[j]);
18     }
19     swap(A[l], A[j]);
20     return j;
21 }
22
23 void quickSort(int A[], int l, int h)
24 {
25     if (l < h)
26     {
27         int pivot = partition(A, l, h);
28         quickSort(A, l, pivot);
29         quickSort(A, pivot + 1, h);
30     }
31 }
32
33 int main()
34 {
35     int n;
36     cin >> n;
37     int arr[n];
38     for (int i = 0; i < n; i++)
39         cin >> arr[i];
40     if (n > 0)
41         quickSort(arr, 0, n - 1);
42     for (int i = 0; i < n; i++)
43         cout << arr[i] << " ";
44     return 0;
45 }
```

```
"C:\Users\Kawsar Hoque\Desl  X  +  v
5
9 8 4 6 11
4 6 8 9 11
Process returned 0 (0x0)   execution time : 14.033 s
Press any key to continue.
```

#### 4) Quick Sort - Pivot as First Element (Using Do-While Loop)

```
Start here x quicksort_first_pivot_for270.cpp x
1  #include <iostream>
2  using namespace std;
3
4  int partition(int A[], int l, int h)
5  {
6      int pivot = A[l];
7      int j = l;
8
9      for (int i = l + 1; i <= h; i++)
10     {
11         if (A[i] < pivot)
12         {
13             j++;
14             swap(A[i], A[j]);
15         }
16     }
17     swap(A[l], A[j]);
18     return j;
19 }
20
21 void quickSort(int A[], int l, int h)
22 {
23     if (l < h)
24     {
25         int pivot = partition(A, l, h);
26         quickSort(A, l, pivot - 1);
27         quickSort(A, pivot + 1, h);
28     }
29 }
30
31 int main()
32 {
33     int n;
34     cin >> n;
35     int arr[n];
36     for (int i = 0; i < n; i++)
37         cin >> arr[i];
38     if (n > 0)
39         quickSort(arr, 0, n - 1);
40     for (int i = 0; i < n; i++)
41         cout << arr[i] << " ";
42     return 0;
43 }
```

```
"C:\Users\Kawsar Hoque\Desl x + v - □ x
5
9 8 7 6 5
5 6 7 8 9
Process returned 0 (0x0)   execution time : 21.705 s
Press any key to continue.
|
```

## 5) Quick Sort - Pivot as Last Element

```
Start here x quicksort_last_pivot270.cpp x
1  #include <iostream>
2  using namespace std;
3
4  int partition(int A[], int l, int h)
5  {
6      int pivot = A[h];
7      int i = l - 1;
8
9      for (int j = l; j < h; j++)
10     {
11         if (A[j] < pivot)
12         {
13             i++;
14             swap(A[i], A[j]);
15         }
16     }
17     swap(A[i + 1], A[h]);
18     return i + 1;
19 }
20
21 void quickSort(int A[], int l, int h)
22 {
23     if (l < h)
24     {
25         int pivot = partition(A, l, h);
26         quickSort(A, l, pivot - 1);
27         quickSort(A, pivot + 1, h);
28     }
29 }
30
31 int main()
32 {
33     int n;
34     cin >> n;
35     int arr[n];
36     for (int i = 0; i < n; i++)
37         cin >> arr[i];
38     if (n > 0)
39         quickSort(arr, 0, n - 1);
40     for (int i = 0; i < n; i++)
41         cout << arr[i] << " ";
42     return 0;
43 }
```

```
"C:\Users\Kawsar Hoque\Desl x + v
8
4 2 2 8 3 3 1 5
1 2 2 3 3 4 5 8
Process returned 0 (0x0)   execution time : 2.652 s
Press any key to continue.
```